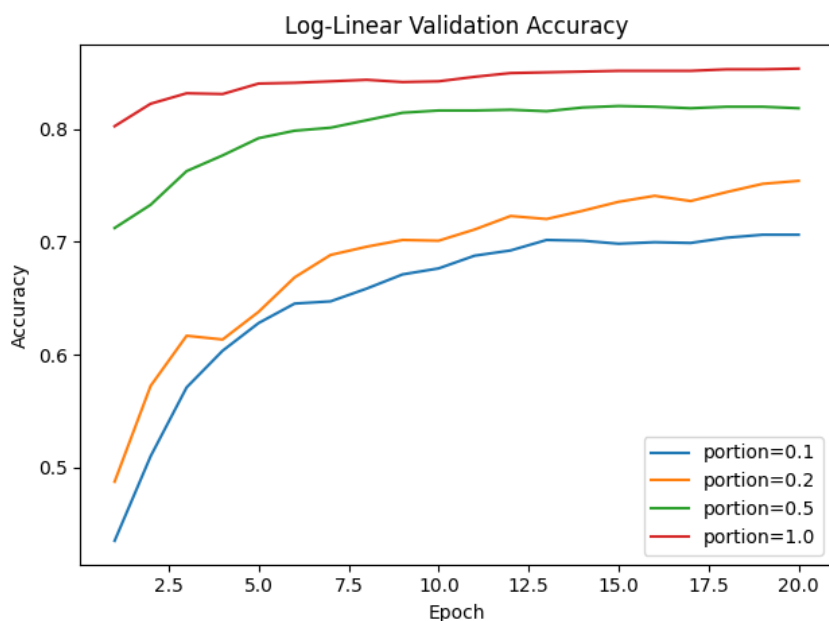
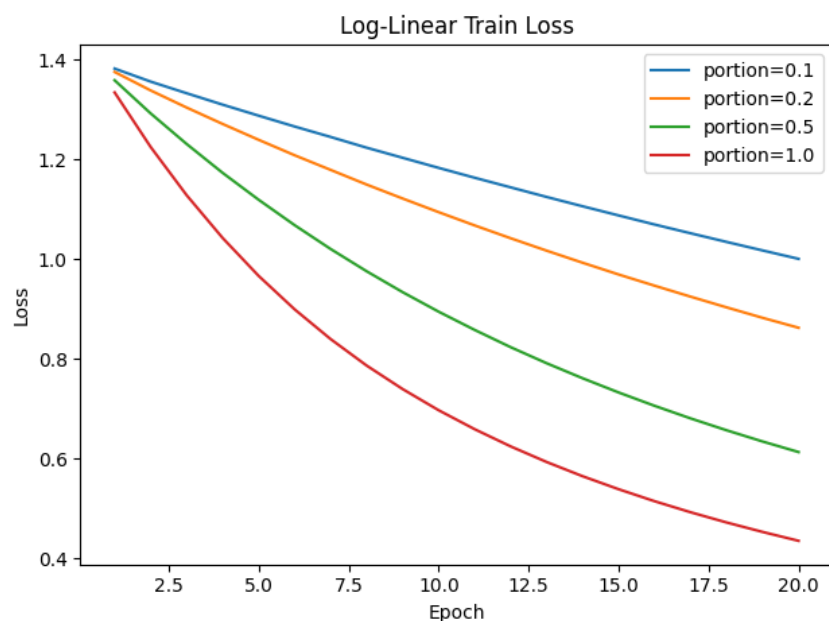


Question 1

Saturday, 2 May 2026 16:55

Here we can see the training loss and accuracy over the validation set, in each of the log-linear models. The only difference between them here is the portion of the data they were trained on. As we'd expect, bigger dataset achieves better loss and higher accuracy, with diminishing returns to some extent. Also, the larger dataset induce a steeper and more stable learning curve.



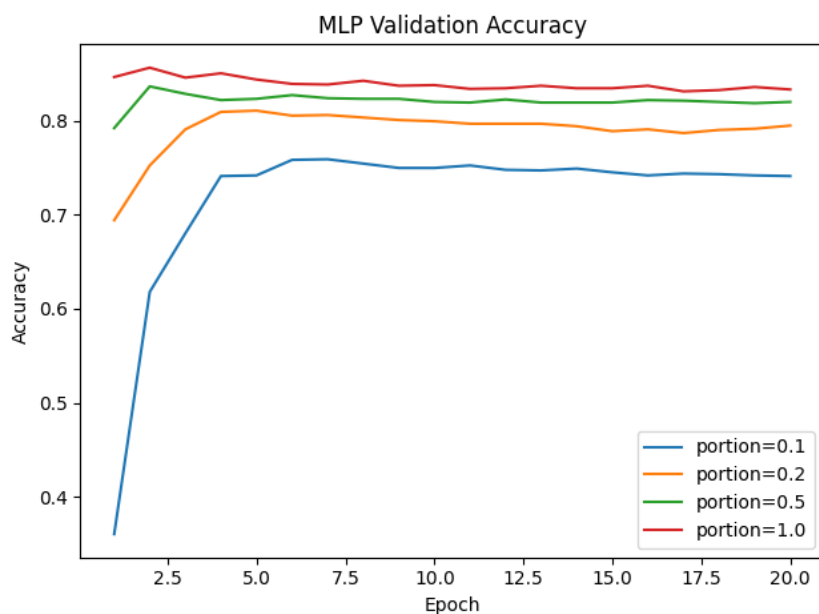
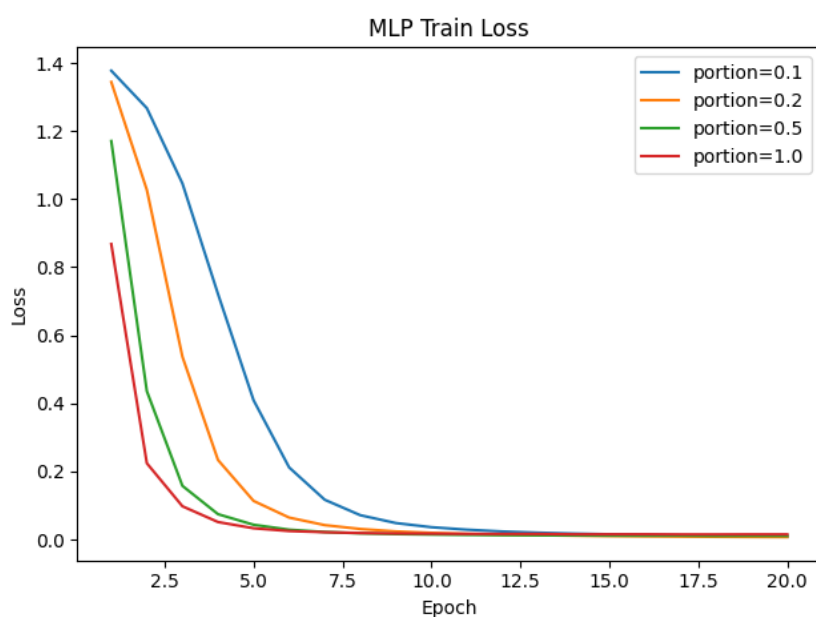
Question 2

Saturday, 2 May 2026 16:58

Here we trained an MLP model. With an additional dense+Relu layer compared to the log-linear one (which had only input -> dense -> softmax).

We can see on the one hand that the additional dense layer allows for much faster training, with all models finishing most of the learning by epoch 8 (the log-linear models' learning curves were much slower).

On the other hand, there are signs indicating overfit - train loss is almost zeros in later epochs, but validation accuracy plateaus at around 0.75 accuracy (depending on train portion).

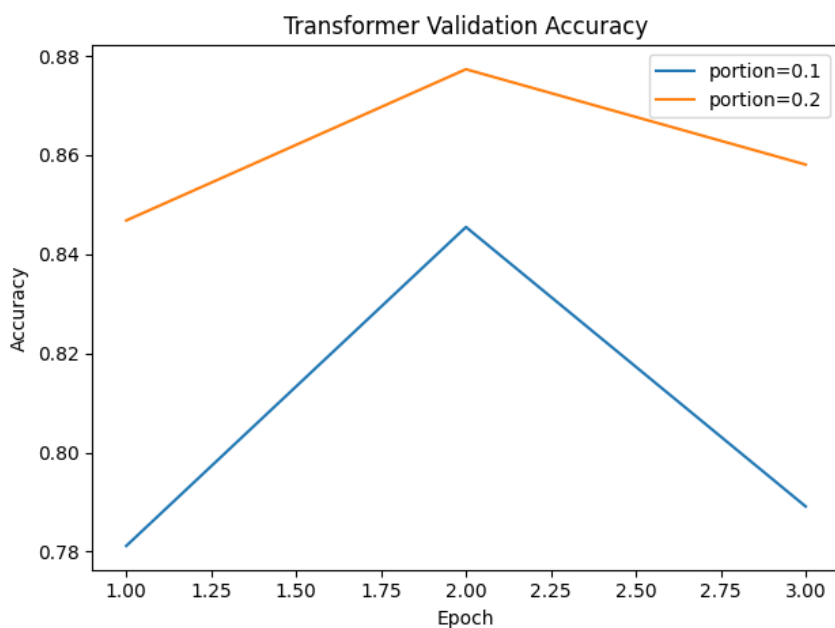
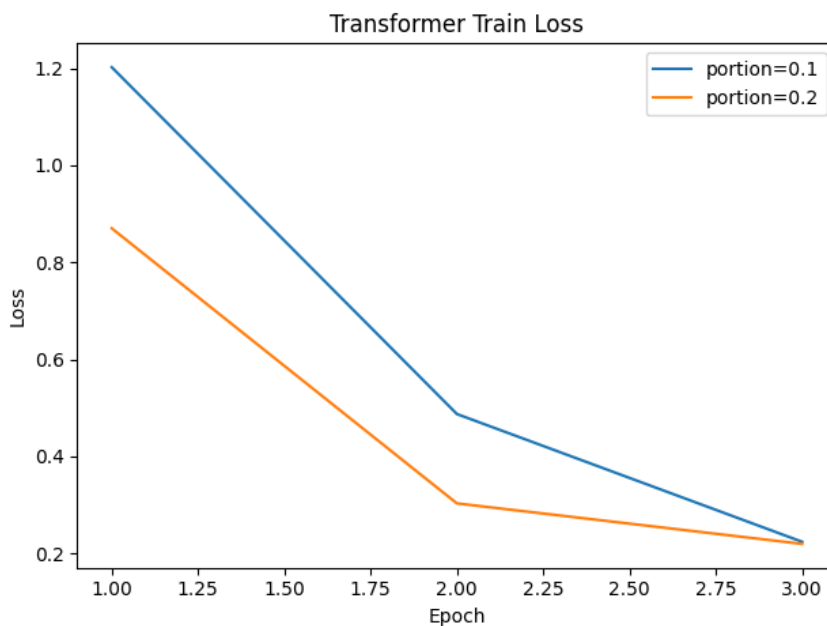


Question 3

Saturday, 2 May 2026 17:45

Here we can see the transformer training process. The train loss achieved by the two models is similar after 2 epochs, indicating quick convergence. This is probably because of the pretraining, that causes the actual training to have less significant effect on model output.

The accuracy of the two models is great overall compared to the other models, with the decrease in the 3rd epoch explained by overfitting - the ratio of the training examples in each epoch (100s) compared to the learned parameters (82M) could cause such behavior.



Question 4

Saturday, 2 May 2026

17:29

Summarizing all 10 models with their accuracy and specific training details:

Model Type	Weights	Percentage of Data	Accuracy percent in final epoch
Log-Linear	8004	10%	70.60%
Log-Linear	8004	20%	75.40%
Log-Linear	8004	50%	81.80%
Log-Linear	8004	100%	85.30%
MLP	1,002,504	10%	74.10%
MLP	1,002,504	20%	79.50%
MLP	1,002,504	50%	82.00%
MLP	1,002,504	100%	83.40%
Transformer (pretrained and distilled)	82,121,476	10%	78.90%
Transformer (pretrained and distilled)	82,121,476	20%	85.80%

- A. We can see the best accuracy with the transformer with 20% of the data (comparable only to the log-linear model on the full dataset).
- B. The log-linear model shows the most sensitivity to the dataset size, at least in the sense that having only 10% of the dataset hurts it most compared to the other models. We could argue that the additional 10% in the jump from 10->20 was most valuable for the transformer that got almost 7% better accuracy from it. The MLP is the least sensitive, with additional data improving accuracy only by 2-3%. That is another indicator of overfit in the MLP model - it already memorized the first 10 percent, so the additional data does not change it's output dramatically.
- C. While generally, the more weights the model has - the better it's accuracy, the increase is strictly non linear. The transformer has almost 5000 times the amount of parameters compared to the log-linear, but shows an accuracy gap of only 7-8% at most. Additionally, the transformer did not go through the same training process as the others - it was pretrained on some related NLP task and then distilled, so the model weights is not a fair comparison here- though it emphasizes even more that more weights are not strictly proportional to better accuracy.